G1 Understanding differentiation

Name: ______________________

Class: ______________________

Date: ______________________

Time: **154 minutes**

Marks: **128 marks**

Comments:

Q1.

(a) (i) Solve the differential equation $\frac{dy}{dt} = y \sin t$ to obtain y in terms of t.

(4)

(ii) Given that $y = 50$ when $t = \pi$, show that $y = 50e^{-(1 + \cos t)}$.

(3)

(b) A wave machine at a leisure pool produces waves. The height of the water, y cm, above a fixed point at time t seconds is given by the differential equation

$$\frac{dy}{dt} = y \sin t$$

(i) Given that this height is 50 cm after π seconds, find, to the nearest centimetre, the heigh tof the water after 6 seconds.

(2)

(ii) Find $\frac{d^2y}{dt^2}$ and hence verify that the water reaches a maximum height after π seconds.

(4)

(Total 13 marks)

Q2.

A model helicopter takes off from a point O at time $t = 0$ and moves vertically so that its height, y cm, above O after time t seconds is given by

$$y = \frac{1}{4}t^4 - 26t^2 + 96t, \qquad 0 \le t \le 4$$

(a) Find:

(i) $\frac{dy}{dt}$;

(3)

(ii) $\frac{d^2y}{dt^2}$

(2)

(b) Verify that y has a stationary value when $t = 2$ and determine whether this stationary value is a maximum value or a minimum value.

(4)

(c) Find the rate of change of y with respect to t when t = 1

(2)

(d) Determine whether the height of the helicopter above O is increasing or decreasing at the instant when t = 3.

(2)

(Total 13 marks)

Q3.

(a) A curve has equation $y = (x^2 - 3)e^x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find $\frac{d^2y}{dx^2}$.

(2)

(b) (i) Find the x-coordinate of each of the stationary points of the curve.

(4)

(ii) Using your answer to part (a)(ii), determine the nature of each of the stationary points.

(2)

(Total 10 marks)

Q4.

A biologist is researching the growth of a certain species of hamster. She proposes that the length, x cm, of a hamster t days after its birth is given by

$$x = 15 - 12e^{-\frac{t}{14}}$$

(a) Use this model to find:

(i) the length of a hamster when it is born;

(1)

(ii) the length of a hamster after 14 days, giving your answer to three significant figures.

(2)

(b) (i) Show that the time for a hamster to grow to 10 cm in length is given by $t = 14\ln\left(\frac{a}{b}\right)$, where a and b are integers.

(3)

(ii) Find this time to the nearest day.

(1)

(c) (i) Show that

$$\frac{dx}{dt} = \frac{1}{14}(15 - x)$$

(3)

(ii) Find the rate of growth of the hamster, in cm per day, when its length is 8 cm.

(1)

(Total 11 marks)

Q5.

The volume, V m^3, of water in a tank at time t seconds is given by

$$V = \frac{1}{3}t^6 - 2t^4 + 3t^2, \quad \text{for } t \geq 0$$

(a) Find:

(i) $\dfrac{dV}{dt}$;

(3)

(ii) $\dfrac{d^2V}{dt^2}$.

(2)

(b) Find the rate of change of the volume of water in the tank, in m^3 s^{-1}, when $t = 2$.

(2)

(c) (i) Verify that V has a stationary value when $t = 1$.

(2)

(ii) Determine whether this is a maximum or minimum value.

(2)

(Total 11 marks)

Q6.

At the point (x, y), where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7$$

(a) (i) Verify that $\dfrac{dy}{dx} = 0$ when $x = 4$.

(1)

(ii) Write $\frac{16}{x^2}$ in the form $16x^k$, where k is an integer.

(1)

(iii) Find $\frac{d^2y}{dx^2}$.

(3)

(iv) Hence determine whether the point where $x = 4$ is a maximum or a minimum, giving a reason for your answer.

(2)

(b) The point $P(1, 8)$ lies on the curve.

(i) Show that the gradient of the curve at the point P is 12.

(1)

(ii) Find an equation of the normal to the curve at P.

(3)

(c) (i) Find $\int (3x^{\frac{1}{2}} + \frac{16}{x^2} - 7)dx$

(3)

(ii) Hence find the equation of the curve which passes through the point P(l, 8).

(3)

(Total 17 marks)

Q7.

(a) A curve has equation $y = e^{2x} - 10e^x + 12x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find $\frac{d^2y}{dx^2}$.

(1)

(b) The points P and Q are the stationary points of the curve.

(i) Show that the x-coordinates of P and Q are given by the solutions of the equation

$$e^{2x} - 5e^x + 6 = 0$$

(1)

(ii) By using the substitution $z = e^x$, or otherwise, show that the x-coordinates of P and Q are $\ln 2$ and $\ln 3$.

(3)

(iii) Find the y-coordinates of P and Q, giving each of your answers in the form $m + 12 \ln n$, where m and n are integers.

(3)

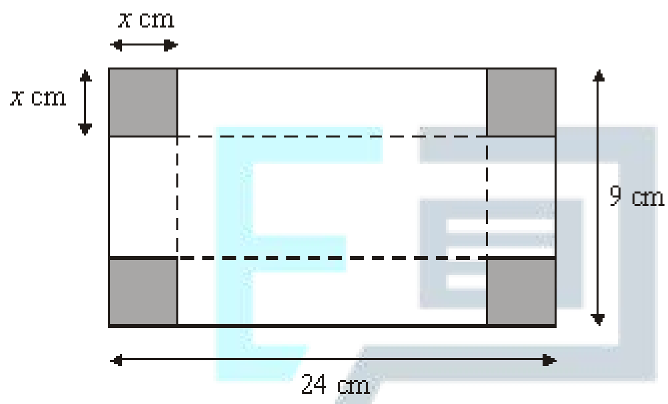
(iv) Using the answer to part (a)(ii), determine the nature of each stationary point.

(3)

(Total 13 marks)

Q8.

The diagram below shows a rectangular sheet of metal 24 cm by 9 cm.



A square of side x cm is cut from each corner and the metal is then folded along the broken lines to make an open box with a rectangular base and height x cm.

(a) Show that the volume, V cm^3, of liquid the box can hold is given by

$$V = 4x^3 - 66x^2 + 216x$$

(3)

(b) (i) Find $\frac{dV}{dx}$.

(3)

(ii) Show that any stationary values of V must occur when $x^2 - 11x + 18 = 0$

(2)

(iii) Solve the equation $x^2 - 11x + 18 = 0$.

(2)

(iv) Explain why there is only one value of x for which V is stationary.

(1)

(c) (i) Find $\frac{d^2V}{dx^2}$.

(2)

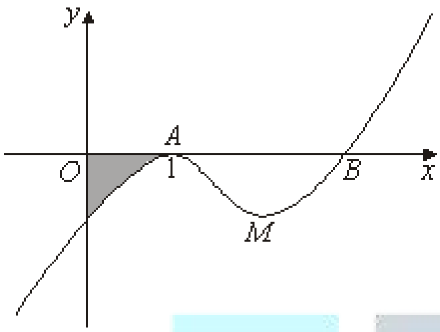
(ii) Hence determine whether the stationary value is a maximum or minimum.

(2)

(Total 15 marks)

Q9.

The curve with equation $y = x^3 - 5x^2 + 7x - 3$ is sketched below.



The curve touches the x-axis at the point A (1, 0) and cuts the x-axis at the point B.

(a) (i) Use the factor theorem to show that $x - 3$ is a factor of

$$p(x) = x^3 - 5x^2 + 7x - 3$$

(2)

(ii) Hence find the coordinates of B.

(1)

(b) The point M, shown on the diagram, is a minimum point of the curve with equation $y = x^3 - 5x^2 + 7x - 3$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Hence determine the x-coordinate of M.

(3)

(c) Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

(2)

(d) (i) Find $\int (x^3 - 5x^2 + 7x - 3)\, dx$.

(4)

(ii) Hence determine the area of the shaded region bounded by the curve and the coordinate axes.

(4)
(Total 18 marks)

Q10.

The function f is defined for all real values of x by

$$f(x) = x^3 + x$$

(a) Express $f(2 + h) - f(2)$ in the form

$$ph + qh^2 + rh^3$$

where p, q and r are integers.

(5)

(b) Use your answer to part (a) to find the value of $f'(2)$.

(2)
(Total 7 marks)